

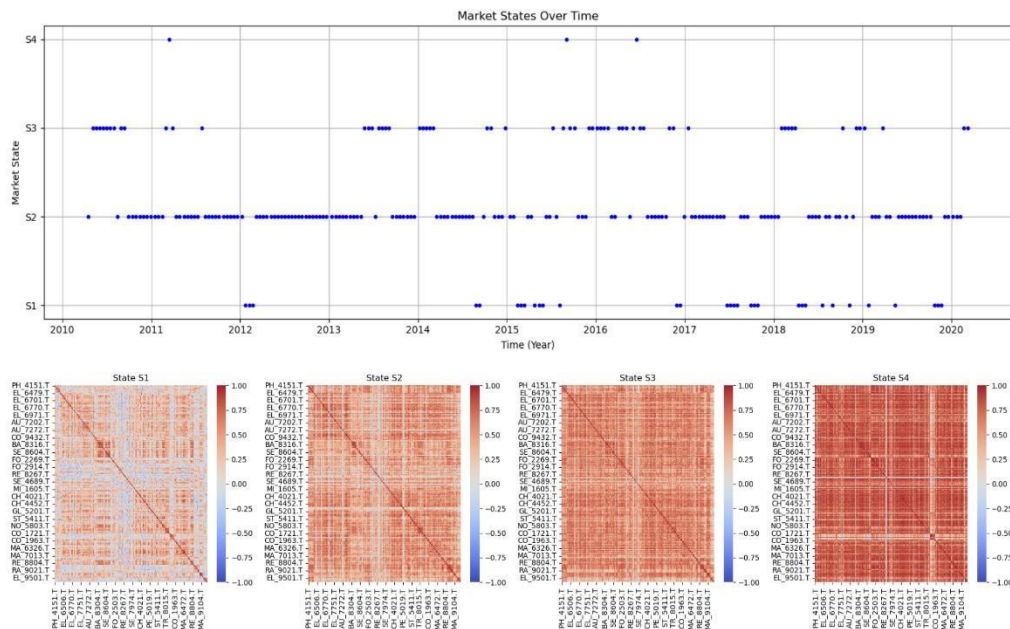
As we begin our journey in 2010, the Nikkei 225 was emerging from the shadows of the global financial crisis. On April 1, 2010, the index opened at 10,228.92, a sign of gradual recovery from its March 2009 low of 7,054.98.

Key Observations:

- The graph shows a predominance of State 2 (S2), likely indicating moderate growth.
- Occasional shifts to S1 suggest periods of caution or minor setbacks.

Market Research Insight: The recovery was fuelled by increased exports and government stimulus measures aimed at boosting consumer confidence and spending.

2. The Great East Japan Earthquake: A Test of Resilience (March 11, 2011).



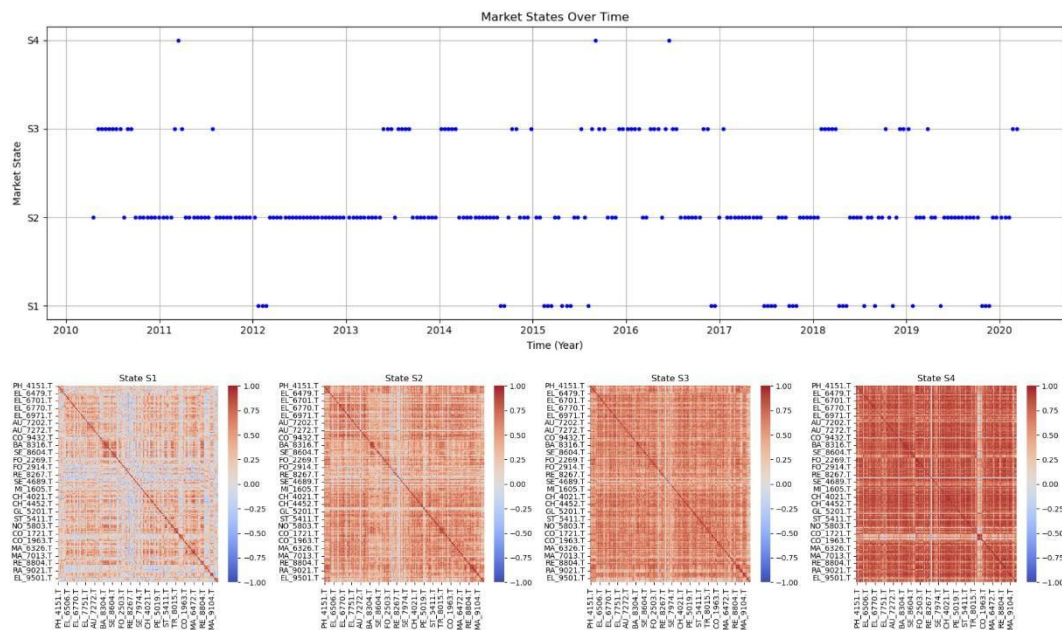
On March 15, 2011, following the devastating earthquake and tsunami, the Nikkei experienced a sharp 10.55% drop, closing at around 8,605.

Key Observations:

- Increased volatility is visible, with more frequent transitions between S1 and S2.
- Despite the shock, the market quickly returned to S2, demonstrating resilience.

Market Research Insight: Analysts noted severe impacts on investor sentiment due to uncertainty about reconstruction costs and potential economic slowdown.

3. The Abenomics Era: A New Hope (2012 onwards)



With the introduction of Abenomics in late 2012, we see a shift in market behaviour.

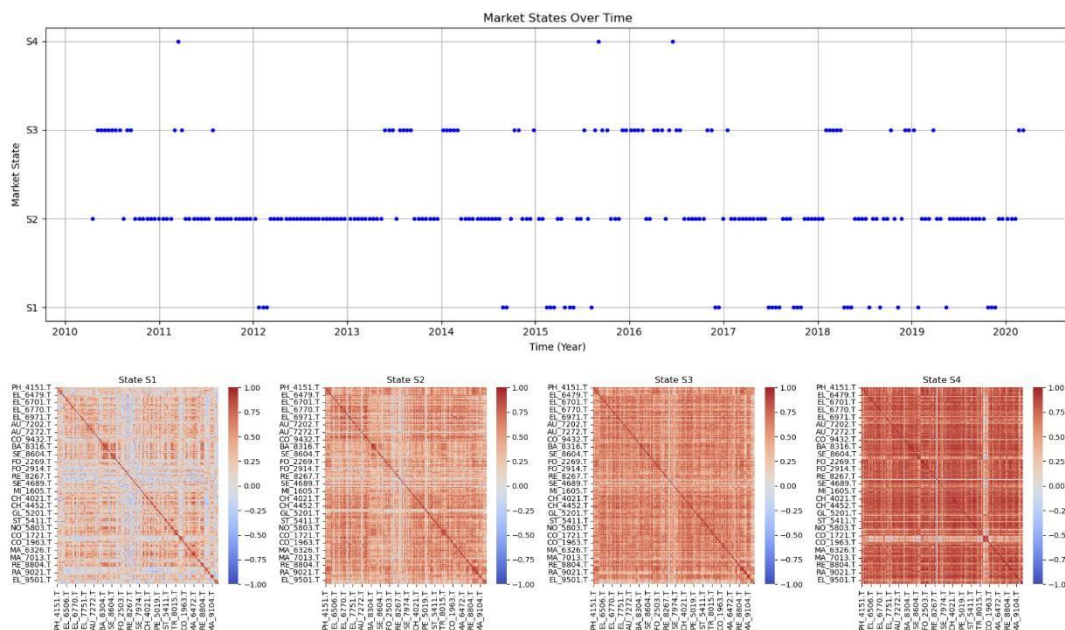
Key Observations:

- More stable and prolonged periods in S2 from 2013 onwards.
- By mid-2013, the Nikkei surged past 15,000, reflecting renewed investor confidence.

Market Research Insight: Abenomics policies, including aggressive monetary easing and fiscal stimulus, were largely successful in driving up stock prices.

4.

Global Concerns and Market Volatility (Early 2016)



Early 2016 saw significant market volatility amid global economic concerns and a strong yen.

Key Observations:

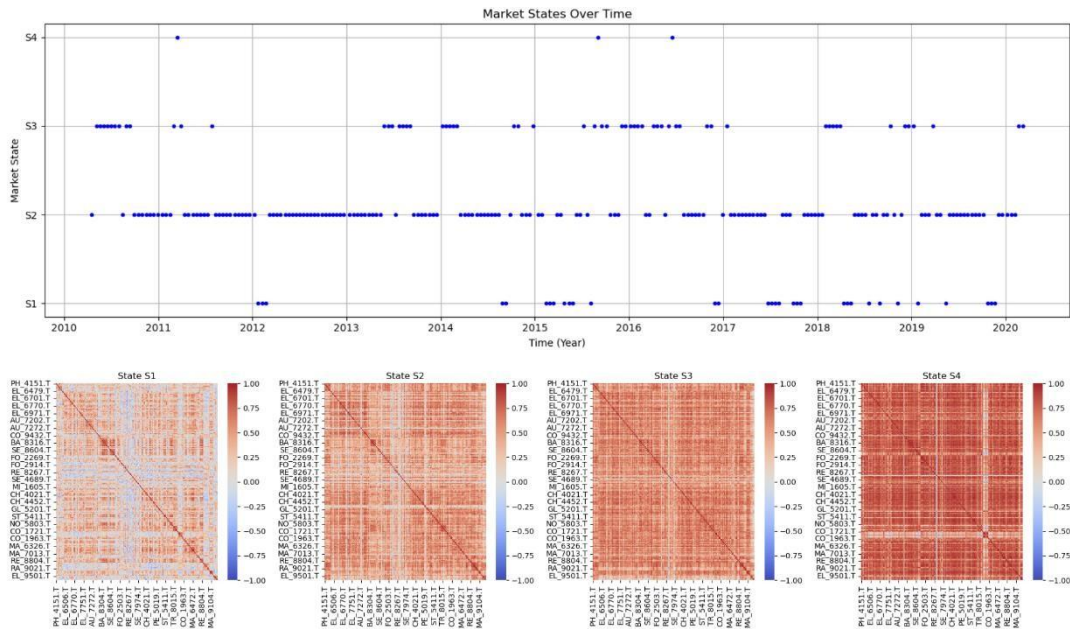
- Increased transitions between S2 and S3, representing higher volatility.
- The Nikkei dropped to around 15,500 in February 2016.

Market Research Insight:

Factors included fluctuating oil prices and fears of a slowdown in China's economy impacting Japanese exports.

5.

The COVID-19 Shock: An Unprecedented Challenge (March 2020)



As our decade-long journey concludes, we witness the beginning of the COVID19 pandemic's impact. **Key Observations:**

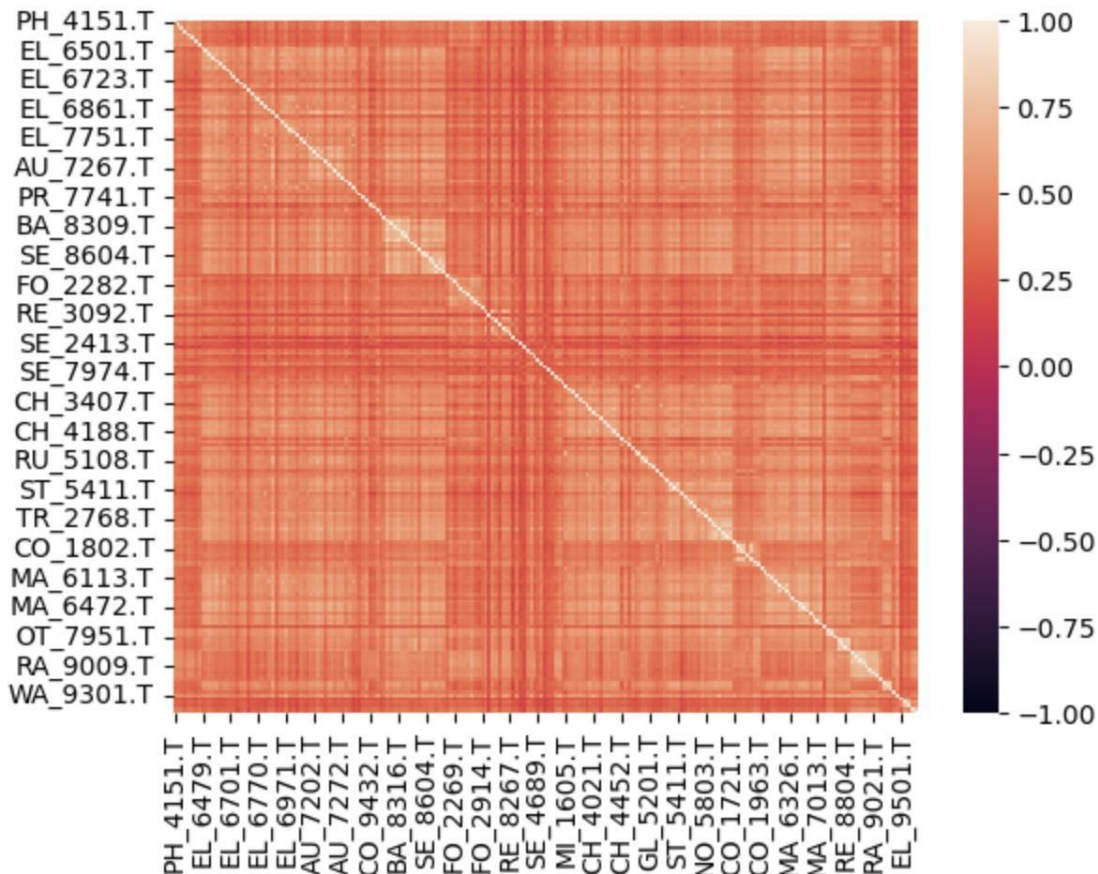
- A rare transition to State 4 (S4) at the very end of the graph.
- By March 23, 2020, the Nikkei had fallen to approximately 16,552.

Market Research Insight:

The pandemic led to widespread economic disruptions, prompting stimulus measures to stabilize the economy.

6.

Understanding Market Dynamics.



This correlation matrix provides insights into the relationships between various financial instruments during our analysis period.

1. Overall Correlation Structure:

- The matrix shows predominantly positive correlations (orange-red colors), indicating that most stocks move together
- The diagonal is naturally white (correlation = 1.0) as it represents self-correlation
- The correlation values range from approximately -1.0 (dark purple) to 1.0 (white)

2. Strong Correlation Clusters:

- There are visible blocks of higher correlation (darker red) among certain groups of stocks:

7.

- The EL (Electronics) sector stocks (EL_6501.T, EL_6723.T, EL_6861.T) show strong correlation
- This could be linked to common factors affecting the electronics industry such as semiconductor cycles or global tech demand

3. Notable Anomalies:

a) Low Correlation Regions:

- Some stocks show relatively lower correlation (lighter red) with others, particularly:
 - FO_2503.T shows generally lower correlation with other stocks
 - This could indicate this company's business model or market segment was less affected by common market factors
 - Possible link to the 2011 Tōhoku earthquake and tsunami which affected different companies differently

b) Negative Correlation Patterns:

- While rare in the matrix, there are some slightly darker patches indicating lower or negative correlations
- This could correspond to:
 - The 2015-2016 Chinese stock market turbulence
 - The 2013 Japanese asset price bubble concerns
 - Different sectors responding differently to Bank of Japan's monetary policies

4. Sector-Based Patterns:

a) Financial Sector Stocks:

- BA (Banking) and IN (Insurance) stocks show strong correlation blocks
- This could be linked to:
 - Bank of Japan's negative interest rate policy introduced in 2016
 - Global financial market conditions post-2008 crisis

b) Manufacturing Sector:

- MA (Manufacturing) stocks show moderate to strong correlations
- Likely influenced by:

8.

- Global supply chain disruptions
- Changes in Japan's industrial production
- Trade tensions with key partners

5. Temporal Implications:

- The strong overall positive correlation suggests that during major market events like:
 - The 2011 Tōhoku earthquake and tsunami.
 - The 2013 Abenomics implementation
 - The 2015-2016 global market volatility
 - The 2019 Japan-South Korea trade dispute Most stocks moved together, indicating market-wide impacts of these events

6. Structural Changes:

- The correlation pattern suggests structural changes in the Japanese market due to:
 - Corporate governance reforms
 - Changes in cross-shareholding practices
 - Increased foreign investor participation

Key Observations:

- Strong positive correlations between many instruments, typical in financial markets.
- Some instruments (e.g., PH_4151.T) show weaker correlations, potentially offering diversification benefits.

Transition Statistics:

S1 to S1: 17 transitions
S1 to S2: 14 transitions
S1 to S3: 1 transitions
S2 to S1: 12 transitions
S2 to S2: 120 transitions
S2 to S3: 20 transitions
S3 to S1: 3 transitions
S3 to S2: 17 transitions
S3 to S3: 34 transitions
S3 to S4: 3 transitions
S4 to S3: 3 transitions

Transition Probabilities:

Probability S1 to S1: 0.531
Probability S1 to S2: 0.438
Probability S1 to S3: 0.031
Probability S2 to S1: 0.079
Probability S2 to S2: 0.789
Probability S2 to S3: 0.132
Probability S3 to S1: 0.053
Probability S3 to S2: 0.298
Probability S3 to S3: 0.596
Probability S3 to S4: 0.053
Probability S4 to S3: 1.000

This data gives us a quantitative understanding of market state transitions.

Key Observations:

- High probabilities for S1 to S1 (0.531), S2 to S2 (0.789), and S3 to S3 (0.596) indicate market momentum.
- Rare transitions to S4 align with major market shocks.

Conclusion:

The Nikkei 225's journey from 2010 to 2020 tells a story of resilience and adaptation. From the aftermath of the global financial crisis, through natural disasters, bold economic policies, and global market turbulence, Japan's key index has demonstrated its ability to recover and grow.

As we look to the future, understanding these patterns and the index's responsiveness to various stimuli will be crucial for investors and policymakers alike. The Nikkei 225's performance reflects not just Japan's economic health, but also its intricate connections to the global financial ecosystem.

This analysis provides a foundation for anticipating future market behaviors and crafting strategies to navigate the complex world of Japanese and global finance.

Sources and Links:

<https://liveindex.org/nikkei/> https://en.wikipedia.org/wiki/Nikkei_225

https://en.wikipedia.org/wiki/Japanese_asset_price_bubble

<https://economictimes.indiatimes.com/markets/stocks/news/japans-nikkei-225-stock-index-sinks-10-in-worst-losses-since-1987/articleshow/112278944.cms>

<https://www.afrugaldocor.com/home/japans-lost-decades-30-years-of-negative-returns-from-the-nikkei-225> <https://www.nytimes.com/2024/02/22/business/japan-stocks-record.html>